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RESEARCH ARTICLE

GD-MVO: Monocular Geometric Visual Odometry Using Image Depth Prediction

ZHUANG LI, JIANJUN CHEN, ABDUL WAHEED, FEI WANG, AND YAN FAN
School of Computer Science, Jiangsu University of Science and Technology, Zhenjiang 212003, China

Corresponding author: Jianjun Chen (jianjunchen@just.edu.cn)

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ABSTRACT Traditional geometric methods estimate camera motion trajectories by analyzing image feature points or pixel information, demonstrating robust performance in certain scenarios. However, these approaches struggle in low-texture and highly dynamic environments. In contrast, end-to-end neural networks can directly predict camera motion trajectories from consecutive image frames, eliminating the need for complex feature extraction and matching processes. Nonetheless, the accuracy of this end-to-end approach still has a significant gap compared to traditional methods. We designed GD-MVO, a monocular visual odometry system that combines the strengths of two approaches. Our system uses both monocular and stereo image pairs to train a deep neural network for depth prediction, then integrates this depth information with geometric features extracted from sequential images to achieve more accurate visual positioning. In this work, we propose 1) a novel integration of geometric methods and deep learning for robust visual odometry; 2) an effective strategy to mitigate the impact of dynamic feature points on prediction outcomes; 3) a fly-out mask technique to resolve the fly-out padding issue in monocular depth prediction networks. Experimental results show that our GD-MVO not only exhibits excellent stability in high-dynamic scenes but also maintains reliable running performance on our specially customized low-texture dataset. In the KITTI odometry benchmark, GD-MVO outperforms both DF-VO and TSformer-VO across all metrics. Most impressively, it achieves a translation error of merely 1.778%, surpassing DF-VO (6.747%) and RAUM-VO (3.676%). Source code available at https://github.com/lilili996/GD-MVO

INDEX TERMS Visual odometry, self-supervised learning, depth prediction, FLANN.

I. INTRODUCTION

Visual Odometry (VO), a critical component in robotic navigation, refers to the process of determining a robot's motion trajectory through the analysis of sequential input images. This technology serves as one of the core modules in simultaneous localization and mapping (SLAM) systems, playing a pivotal role in enabling autonomous navigation in various environments.

In recent years, visual odometry utilizing a stereo camera has achieved remarkable success. The popularity of stereoscopic visual odometry stems from its ability to estimate reliable depth maps, which are crucial for computing accurate transformation matrices. This capability allows for robust trajectory estimation in diverse scenarios, from indoor robotics to autonomous vehicles. However, stereoscopic systems face a significant limitation: when the distance between the camera and the observed scene substantially exceeds the camera baseline, the stereo setup effectively degenerates into a monocular system [1]. This phenomenon, known as the baseline degeneracy problem, can severely compromise the accuracy of depth estimation and, consequently, the overall trajectory measurement.

In contrast, visual odometry systems employing monocular cameras are inherently immune to this baseline-related issue. Traditional monocular visual odometry relies on geometric principles, integrating multi-view geometric constraints and feature tracking techniques to estimate the camera's motion trajectory. While this approach performs adequately in controlled environments, it is susceptible to failure in scenes characterized by limited texture, dynamic objects, or variable illumination [2]. Furthermore, the inherent scale ambiguity

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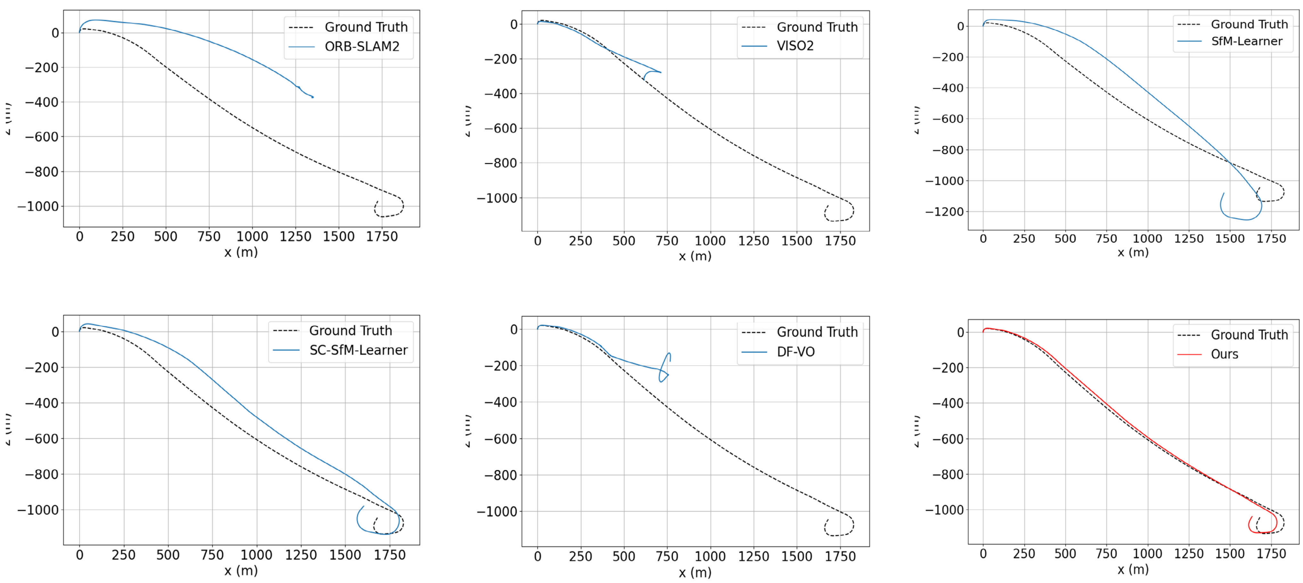


FIGURE 1. **Trajectory performance of some classical visual odometry in KITTI 01.**

in monocular odometry necessitates additional assumptions or constraints to rectify the scale factor, potentially leading to prolonged periods of drift and inconsistency. To address these challenges, a novel monocular visual odometry system is proposed in this paper.

This innovative approach incorporates a single-view depth prediction network to estimate the depth of monocular images and leverages this depth information to correct the scale factor. Traditional visual odometry is based on the assumption of a static scene. This means that the objects in the observed scene do not change over time. However, real-world environments frequently contain dynamic elements such as pedestrians and vehicles. These moving objects can introduce erroneous pixel variations, potentially compromising the accuracy of the odometry system. To enhance the visual odometer's robustness in dynamic environments, a dual-solver system complemented by a scale factor correction module has been developed in this paper. The approach dynamically switches between two operational modes. When the scale factor correction module functions optimally, the 2D Attitude Solver is utilized to track the camera's motion trajectory. In instances where the scale factor correction module fails, the system seamlessly transitions to the 3D Attitude Solver for robust pose estimation. This adaptive strategy enables the system to maintain high accuracy across a diverse range of scenarios, effectively mitigating the impact of dynamic objects on the odometry calculations.

In summary, the main contributions of this paper can be listed as follows.

- GD-MVO merges the traditional geometric method with deep learning-based depth prediction, resulting in a novel hybrid approach. In the feature extraction and matching stage, we adopt the feature point method with stronger robustness to illumination changes and viewpoint variations, instead of the DF-VO optical flow prediction scheme or the CNN-SVO semi-direct method, which are also hybrid methods. This integration improves the algorithm's generalization capabilities, enabling robust performance across diverse environments.
- GD-MVO incorporates a dual-solver system, consisting of a 2D Attitude Solver and a 3D Attitude Solver, which work in tandem with a scale factor correction module. This allows our method to adaptively switch between the two solvers based on the reliability of the scale factor correction, thereby enhancing the system's robustness in dynamic environments. This dual solver approach is a key innovation that sets our work apart from existing hybrid methods.
- We address the problem of persistent flyout padding inherent in depth prediction networks by implementing a flyout mask. This effectively reduces the interference of depth prediction errors at image boundaries on attitude estimation, thereby improving the overall accuracy of the visual odometry.

The remainder of this paper is organized as follows: Section II provides an overview of related work in the field of monocular visual odometry. Section III introduces our proposed GD-MVO system, detailing its key components. The experimental setup, results, and comparative analysis are presented in Section IV. Finally, Section V summarizes the key contributions of this work and discusses potential future directions in the field of monocular visual odometry.

II. RELATED WORK

A. GEOMETRY-BASED VISUAL ODOMETRY

The foundations of geometry-based VO can be traced back to pioneering work in multi-view geometry and Structure

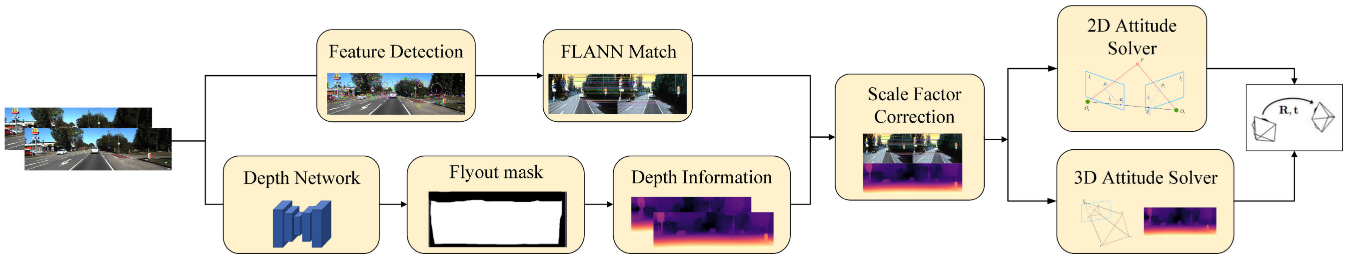


FIGURE 2. Architecture of GD-MVO.

from Motion (SfM), evolving from classical computer vision research. These approaches rely on a sophisticated pipeline encompassing feature detection, description, matching, and pose estimation. This pipeline leverages advanced techniques such as corner and blob detectors, coupled with distinctive descriptors like SIFT and ORB, and incorporates geometric constraints including epipolar geometry [3], [4] and PnP methods [5]. The practical implementation of these principles has led to the development of influential VO and Simultaneous Localization and Mapping (SLAM) systems.

Notable examples include ORB-SLAM2 [6], which distinguishes itself through efficient map reuse and loop closing mechanisms; Direct Sparse Odometry [7], an innovative system that synergizes direct methods with sparse approaches. While these methods have demonstrated remarkable accuracy in controlled environments, they face significant challenges when deployed in complex, real-world scenarios. These limitations stem from their reliance on sparse features, assumptions of static scenes and Lambertian surfaces, and inherent issues such as scale ambiguity in monocular systems.

B. END-TO-END VISUAL ODOMETRY BASED ON DEEP LEARNING

With the advent of deep learning techniques, researchers have explored data-driven methods for VO and depth estimation. Early research focused on supervised learning, utilizing ground truth data for tasks such as relative attitude estimation [8], [9] and depth prediction [10]. However, the scarcity and cost of obtaining accurate ground truth labels have driven the development of self-supervised techniques. Subsequent studies, such as SC-SfM-Learner [11] and Depth-VO-Feat [12], addressed scale inconsistency and dynamic object processing, respectively.

Recently, transformer architecture has attracted much attention for its excellent performance in sequence modeling tasks. Inspired by this, some studies have attempted to use transformer in visual odometry. Zhao et al. proposed a transformer-based self-supervised monocular depth and VO framework (TSSM-VO) [13], which utilizes a multi-head self-attention mechanism to efficiently capture long-range spatial and temporal correlations. Françani and Maximo considered visual odometry as a video understanding problem and proposed a transformer based TSformer-VO model [14], which is capable of extracting spatial-temporal features

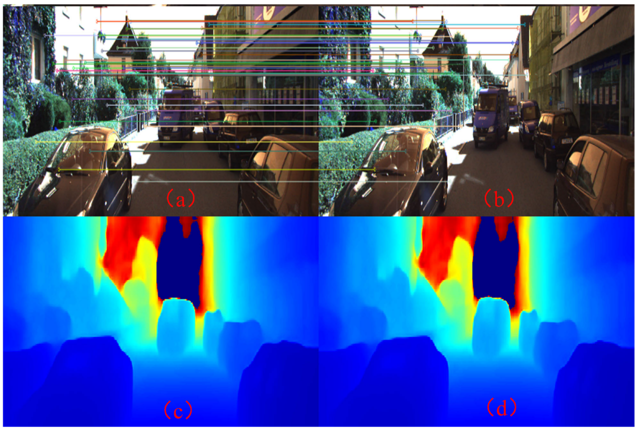


FIGURE 3. (a) and (b) are the reference frame and the current frame, respectively, which are used as inputs to the system. (c) is the depth information of the reference frame for the 3D Attitude Solver. (d) is the depth information of the current frame for scale factor correction.

from video sequences through a self-attention mechanism to directly estimate the 6-degree-of-freedom end-to-end camera pose. Kurt et al. proposed a visual-inertial fusion transformer (VIFT) [15], a causal transformer architecture that utilizes Inertial Measurement Unit (IMU) data for in-depth vision inertial odometry. VIFT utilizes a transformer layer to fuse and distill latent representations from the vision encoder and inertial encoder, allowing for efficient integration of visual and inertial information.

C. HYBRID VISUAL ODOMETRY

Recognizing the complementary nature of geometric and learning-based approaches, researchers have proposed hybrid frameworks that integrate these two approaches. CNN-SLAM [16] and CNN-SVO [17] incorporate single view depth prediction of convolutional neural networks into direct and semi-direct SLAM frameworks, respectively. DF-VO [18] and Yang et al. [19], utilize depth optical flow estimation and depth prediction to establish high-quality correspondences, which are then fed into a geometric pose solver. Cimarelli et al. proposed RAUM-VO [20], which uses Superpoint and Superglue for feature matching and achieves comparable accuracy to more sophisticated hybrid methods while maintaining computational efficiency, effectively solving the common limitations of rotational estimation in unsupervised attitude networks. Cho and Kim [21] propose a novel hybrid visual odometry approach that integrates semantic segmentation, optical flow estimation, and multi view geometry to robustly estimate camera motion in dynamic environments.

These hybrid approaches aim to combine the robustness and accuracy of multi-view geometry with the rich scene-understanding capabilities of deep neural networks. Despite the above efforts, monocular VO still has some challenges, including robust scale estimation and handling high-dynamic scenes. Taking the high-dynamic scene as an example, as shown in Fig. 1, most of the classical methods do not perform well in sequence 01 of the KITTI dataset. Aiming to address these challenges, this paper proposes a novel hybrid framework that synergistically combines the advantages of deep learning and multi-view geometry.

III. GD-MVO

The traditional geometry-based visual mileage computation method usually includes extracting image features, feature matching, and estimating camera pose based on feature correspondences. Our proposed GD-MVO introduces depth prediction and 3D Attitude Solver based on the above fundamental processing, as shown in Fig. 2 and Fig. 3. The details of GD-MVO are described in the following subsections.

A. FEATURE DETECTION AND MATCHING

In the feature detection stage, a classical algorithm is adopted to detect and describe the local features of an image: scale-invariant feature transform (SIFT). SIFT was proposed by David Lowe in 1999 [22] and has the advantages of scale invariance, rotation invariance, etc. This algorithm mainly consists of the following four computational steps: (1) scale-space extremum detection, (2) precise keypoint localization, (3) keypoint orientation determination, and (4) keypoint descriptor generation.

For the input images consisting of a reference frame I_i and the current frame I_j, the SIFT algorithm is employed to extract their respective keypoint descriptors. Once these descriptors are obtained for both frames, the next step involves matching the corresponding feature point pairs between I_i and I_j. To perform this feature point matching efficiently, Fast Library for Approximate Nearest Neighbors(FLANN) [23] is utilized. FLANN is a fast approximate nearest neighbor search algorithm that significantly reduces the computational complexity of feature matching. The core principle of FLANN involves constructing a data index structure, such as a K-D tree or a K-Means tree, in a high-dimensional feature space. Subsequently, it performs a nearest-neighbor search on the target feature vector using this structure. By employing FLANN, the number of feature comparisons is greatly reduced, leading to a dramatic decrease in the computational load of feature matching. This efficient process yields the matching results of the corresponding feature pairs $\left(F_i, F_j\right)$.

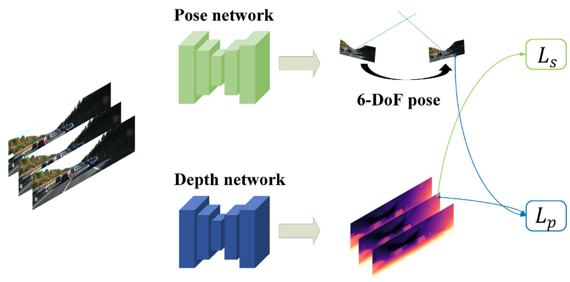


FIGURE 4. Training architecture.

B. SELF-SUPERVISED TRAINING

This section uses a self-supervised approach to jointly train the pose and depth estimation networks, as depicted in Figure 4. The use of the pose network to estimate the camera pose is only necessary during training to help constrain the depth estimation network.

The depth network, based on U-Net [24], adopts an encoder-decoder architecture. It utilizes a ResNet50 feature extractor as its encoder. This network processes a single image input and generates a corresponding dense depth map. Similarly, the pose network incorporates a ResNet50 as its backbone. It takes a pair of images as input and outputs their relative 6-DoF (Degrees of Freedom) pose. For training the depth network, this paper primarily follows the methodology proposed in [25]. The training data consists of a hybrid sequence, incorporating both monocular and stereo data.

Given two image frames, denoted as I_t and I_{t+1}, the depth network generates depth information D_t and D_{t+1} for each frame. Concurrently, the pose network produces the 6-DoF relative pose transformation $T_{t+1\to t}$ between the two frames. Utilizing this information, it becomes possible to warp view I_{t+1} to reconstruct a new view $I_{t+1\to t}$. This reconstruction is achieved through Equation (1).

$$I_{t+1\to t} = w(I_{t+1}, KT_{t+1\to t}K^{-1}xD_{t+1}(x, y)) \tag{1}$$

where $w(\cdot)$ is a differentiable image warping function [26], K is camera intrinsic, (x, y) is the 2D pixel coordinates in the plane of the image I_t.

The photometric error $pe\,(I_{t+1\to t}, I_t)$ between the reconstructed image $I_{t+1\to t}$ and the reference image I_t can be calculated by Equation (2).

$$\begin{aligned} & pe\,(I_{t+1\to t}, I_t) \\ & = \alpha * SSIM\,(I_{t+1\to t}, I_t) + (1-\alpha) * ||I_{t+1\to t} - I_t||_1 \end{aligned} \tag{2}$$

where SSIM is a metric that measures the similarity between two images [27]. $||.||_1$ is L1-norm. In this case, the α value of 0.85 is utilized to balance the SSIM loss and L1 loss.

Specifically, for the reference view I_t, images $I_{t-1\to t}$ and $I_{t+1\to t}$ can be reconstructed based on its neighborhood views I_{t-1} and I_{t+1} respectively. The final photometric loss is computed by Equation (3). Here, while calculating the photometric error between the reference pixel in I_t and the synthesized pixel in $I_{t-1\to t}$ and $I_{t+1\to t}$, only the pair of reference pixel and the synthesized pixel with the smallest error is

FIGURE 5. When the camera is in forward motion, using the I_{t+1} Reverse Warp Reconstruction $I_{t+1\to t}$ would result in edge fly-out, and we use a fly-out mask M_d Perform filtering.

chosen as the photometric error. A smaller photometric error means that the reconstructed view is more accurate, which in turn means that the depth information generated by the depth network is more accurate.

$$L_p = \min\left(pe\left(I_t, I_{t-1\to t}\right), pe\left(I_t, I_{t+1\to t}\right)\right) \tag{3}$$

Following the recommendations set forth by [28], we advocate for the implementation of local smoothing techniques in depth estimation. This is achieved by introducing an L1 penalty into the depth gradient, which is weighted using the image gradient, as shown in Equation (4). This approach is particularly effective in addressing depth discontinuities, which often manifest at the image gradient.

$$L_s(D_t, I_t) = |\partial_x D_t| e^{-|\partial_x I_t|} + |\partial_y D_t| e^{-|\partial_y I_t|} \tag{4}$$

where $\partial_x D_t$ and $\partial_y D_t$ are the gradients of depth information D_t in horizontal and vertical directions respectively; $e^{-|\partial_x I_t|}$ and $e^{-|\partial_y I_t|}$ are weight terms based on the gradient of the image in the current view.

It is observed that from Equation (4) when the magnitude of the image gradient is high, the weight term is low, which permits discontinuities in the depth map. Conversely, in areas with a low image gradient, the weight term is high, resulting in a smooth depth map. The motivation behind incorporating the depth smoothing term is based on the anticipation that the projected depth map will exhibit segmentation in the edge area of the object while maintaining smoothness in the interior region of the object. This behavior aligns with the typical attributes observed in a real-life scene. The inclusion of this loss factor L_s improves the preservation of edges and the smoothing of depth prediction, resulting in more accurate depth estimation outcomes.

We combine the Photometric loss and the depth smoothing loss to obtain the final training loss: $L = L_p + \lambda L_s$. The PyTorch [29] framework is utilized for implementing and training depth network. The training phase utilized the Adam [30] optimization method, with a total of 15 epochs of repetitions and a learning rate of 0.0001. Each batch consists of 12 samples, and the input and output images have a resolution of 640 × 192. The weight coefficient λ for the regular term that smooths the depth while preserving edge information is set to 0.001. The model undergoes training and inference on a high-performance hardware platform equipped with a 12th-generation Intel Core i9 processor, 24GB of video RAM, and an RTX 3090 Ti graphics card. After the training is completed, a depth network is obtained, which is used to predict the depth information of the image. The depth information obtained is used later for scale factor correction and 3D Attitude Solver.

Algorithm 1 2D Attitude Solver

```
Input: 2D-2D match formed between reference view I_j and current view I_i: (F_i, F_j)
Output: relative pose: T_{i→j} = [R, t]
Calculate the average optical flow between (F_i, F_j): F_{i→j}
if F_{i→j} > F_min
    Random reordering of F_i, F_j
    Use RANSAC to robustly estimate the Essential matrix E
Recover [R, t] from the Matrix E and compute the number of
        interior points C_{i→j}
    if C_{i→j} > C_min
        T_{i→j} = [R, t]
    else
        T_{i→j} = 0
else
        T_{i→j} = 0
end if
```

C. CORRECTION FOR SCALE FACTOR

In the training process of depth network with I_{t+1} reverse warp reconstruction $I_{t+1\to t}$ (see Section III-B for details), some pixel coordinates in I_{t+1} may be mapped to coordinates outside the I_t boundary. At this point, the edges of the warped image will fly out, as shown in Fig. 5. Experiments have shown that the inclusion of these pixels in the loss reduces the performance of depth prediction, which has a considerable impact on both the scale factor correction and the 3D Attitude Solver. One option to overcome edge fly-out is to use a mask [25], [31], [32] and only compute the loss in the masked region. However, this approach is ineffective and often results in edge artifacts shadowing [33] in the depth image. Another option is to directly apply border filling to the fly-out region [25], but the depth network trained under this scheme cannot accurately predict the depth information of the image edges in the inference stage.

In this paper, we adopt the strategy of reflection filling followed by a fly-out mask to overcome the edge fly-out problem. During the training phase of the depth network, the flyout portion of the image boundary is filled with reflections, which produces a smoother and more realistic transition at the boundary than boundary filling. As suggested by [34], In the depth image prediction stage, we use a fly-out mask to shield pixels of edge area with lower accuracy, allowing subsequent scale factor correction and 3D Attitude Solver to focus on more reliable regions of the depth map.

Specifically, suppose we have two image frames I_t and I_{t+1}, with corresponding depth maps D_t and D_{t+1}, camera intrinsic parameters K, and inter-frame motion $T_{t\to t+1}$. We first back-project the pixel coordinates $p_t(x, y)$ from I_t to a 3D point P_t in space using the depth value $D_t(x, y)$. The calculation formula is:

$$P_t = D_t(x, y) \cdot K^{-1} \cdot (x, y, 1)^T \tag{5}$$

Then, we transform P_t through the rigid body transformation matrix $T_{t\to t+1}$ to obtain the corresponding 3D point P_{t+1}:

$$P_{t+1}(x, y, z) = T_{t\to t+1} \cdot P_t \tag{6}$$

Next, we project P_{t+1} through the camera intrinsic parameters K onto the pixel plane at time $t + 1$ to obtain its corresponding pixel coordinates $p_{t+1}(x', y') = K \cdot P_{t+1}$.

If the projected pixel coordinates p_{t+1} fall within the valid range of image I_{t+1} (i.e., coordinate values are greater than 0 and less than the image dimensions), it indicates that the point is visible at time $t + 1$, and the fly-out mask $M(x, y)$ is set to 1; otherwise, it is set to 0.

Monocular visual odometry inherently lacks direct access to real-world scale information. Consequently, additional reference data, such as inertial measurement unit (IMU) readings, is typically required to rectify the scale factor [35]. In our approach, we leverage the depth network trained in the preceding section to predict image depth information, which serves as the necessary reference data.

After using the depth network to obtain the depth information D_j of the reference view I_j, a flyout mask M is applied on top of it to obtain the scale recovery reference data D_j^m. The feature points F_i, F_j are normalized to obtain x_i and x_j, and then the 3D point cloud is reconstructed by triangulation.

$$X_j = \mu \begin{bmatrix} x_j \\ 1 \end{bmatrix}, \mu = \frac{1}{(x_i - x_j)^T e} \tag{7}$$

where $e = [0, 0, 1]^T$. The 3D point cloud X_j is projected back to view I_j to obtain a triangulation-based predictive depth information D_j^{tri}.

$$D_j^{tri} = \pi(\mathrm{X}_j) = \mathrm{K}\cdot\mathrm{X}_j \tag{8}$$

where the K is camera intrinsic, a known parameter, $\pi\ (\cdot)$ denotes the projection function.

Depth ratio r_i with the scale factor of the visual odometry s should theoretically satisfy the linear relationship [36]. To robustly estimate this scale factor, we introduced the RANSAC algorithm [37] to fit the noisy data.

$$s = \underset{s}{argmin} \sum\nolimits_i ||\, |r_i - s|\, ||_0 \quad \text{subject to} |r_i - s| \le \varepsilon \tag{9}$$

where $r_i = D_j^{tri} \big/ D_j^m$, $\boldsymbol{\varepsilon}$ is the appropriate residual threshold.

Algorithm 2 3D Attitude Solver

Input: 2D keypoints F_i and depth information D_i^m of current view I_i, 2D keypoints F_j of reference view I_j
Output: relative pose: $T_{i\to j}$
1: Filter F_i to remove the points that are beyond the image boundary and depth illegal values.
2: Calculate the 3D point cloud coordinates X_i by back-projection using the camera's intrinsic and depth information D_i^m.
3: Randomly disrupt the point sequences of F_j and X_i.
4: Estimate $T_{i\to j}$ from 3D-2D corresponding point pairs based on the RANSAC framework.
5: Randomly sample the minimum inner point set and compute the initial solution
6: Using the initial solution as a starting point, perform a local nonlinear optimization to obtain the optimized solution.
7: Determine the inner point set by calculating the point reprojection error from the optimized solution.
8: Repeat 5-6 until the best solution is found.

D. 2D ATTITUDE SOLVER

The primary approach for determining the relative pose between two frames of an image is to solve the essential matrix. The implementation requires the use of our 2D Attitude Solver, as shown in algorithm 1. This involves establishing 2D-2D correspondences between pixels in the two frames, denoted as F_i and F_j (derived from feature detection and matching in Section III-A). The relationship between these points is fulfilled in the following manner.

$$F_j^{\mathrm{T}} K^{-T} E K^{-1} F_i = 0 E = [t]_\times R \tag{10}$$

where R and t are the relative rotation and translation (up to scale) between the two frames, and $[\cdot]_\times$ denotes the skew-symmetric matrix representation of the cross product.

We first calculate the average optical flow magnitude between F_i and F_j. If the magnitude exceeds F_{min}, we proceed to estimate the pose, where F_{min} is an empirical value that we usually set to 10. This adaptive strategy ensure that there is sufficient disparity between the frames for reliable pose estimation. Next, we randomly shuffle the order of correspondences and apply RANSAC to estimate E. In each iteration, we sample a minimal set of 8 correspondences and compute E using the normalized 8 point algorithm. We then recover the relative pose R, t from E via singular value decomposition and calculate the number of inliers $C_{i\to j}$ that have reprojection errors below a certain threshold.

The above steps are repeated for a fixed number of iterations, and the pose with the highest inlier count is chosen as the output of the 2D Attitude Solver. If the inlier count $C_{i\to j}$ is below a threshold C_{min} (Equal to 5% of the number of F_i), we consider the pose estimation unsuccessful and simply set the relative pose to identity, i.e., $T_{i\to j} = 0$.

It's worth noting that the 2D Attitude Solver alone cannot fully recover the scale of the translation. To resolve the scale

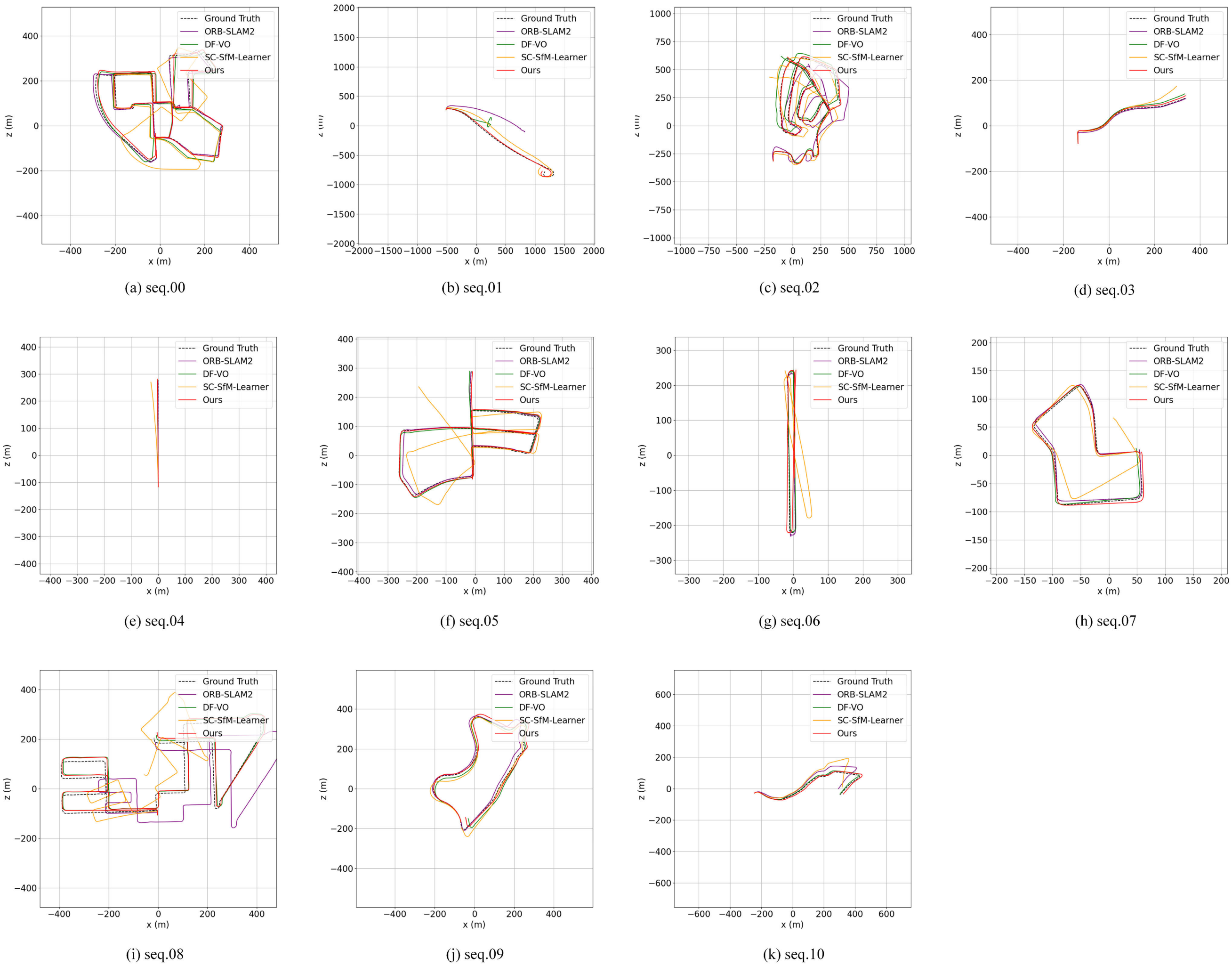


FIGURE 6. **Comparison of trajectories in sequences 00-10.**

ambiguity, we incorporate the scale factor correction module (Section III-C) to obtain the final relative pose: $T_{i\to j} = \{R, s \cdot t\}$, where s is the estimated scale factor.

E. 3D ATTITUDE SOLVER

When the motion of the scene is small, third-party moving objects that break the assumptions of a static scene can greatly interfere with the computation of the 2D Attitude Solver (e.g., when a vehicle is waiting for a traffic light at an intersection), and the Scale Factor Module is rendered ineffective. So we design the 3D Attitude Solver in this subsection, as shown in algorithm 2.

When a set of three-dimensional points and their corresponding image points on a two-dimensional image plane are known, estimating the camera's outer parameters is a classical Perspective-n-Point problem [38]. Traditional PnP methods, like P3P and EPnP, are more susceptible to noise and outliers. In this work, we seamlessly integrate local nonlinear optimization with the RANSAC technique. The approach is capable of achieving precise position estimation results and also exhibits excellent resistance to outliers. Local nonlinear optimization employs a reduced number of interior points for refinement, whereas RANSAC is employed to effectively identify the set of interior points from a vast amount of data.

Following the processing described in Sec.III-C, we initially apply a fly-out mask M to the depth information to acquire the processed valid depth data. We then compute the 3D point cloud coordinates corresponding to the current view by back projection using the camera intrinsic matrix and these practical depth values.

Next, we robustly estimate the initial camera pose from the 3D-2D correspondences. In each RANSAC iteration, we randomly sample a minimal set of correspondences and compute the initial pose $\{R_0, t_0\}$ by minimizing the reprojection error:

$$\{R_0, t_0\} = \arg\min_{R,t} \sum_{i=1}^{n} \rho\left(||F_j - \pi(R, t, X_i)||^2\right) \quad (11)$$

where, ρ denotes a robust kernel function, such as truncated quadratic;F_j denotes a 2D-pixel coordinate in the reference view; X_i denotes the corresponding 3D spatial point coordinate in the current view; and π represents a projection function, which projects a 3D point onto a 2D pixel plane.

Starting from the RANSAC estimate $\{R_0, t_0\}$, we further refine the camera pose by performing local nonlinear optimization over all the 3D-2D correspondences:

$$\{R^*, t^*\} = \arg\min_{R,t} \sum\nolimits_{i=1}^{n} ||F_j - \pi(R, t, X_i)||^2 \tag{12}$$

The optimization objective is to minimize the sum of squared reprojection errors. We solve this nonlinear least-squares problem using the Levenberg-Marquardt algorithm. The optimized pose $\{R^*, t^*\}$ is then evaluated on all the correspondences to compute the number of inliers that have reprojection errors below a certain threshold ϵ:

$$I = \{i|\ ||F_j - \pi(R^*, t^*, X_i)|| < \epsilon\} \tag{13}$$

Continue executing the aforementioned procedure until either the maximum number of iterations is reached or the number of interior points I ceases to rise, resulting in the attainment of the optimal solution $[R, t]$.

IV. EXPERIMENT

This section demonstrates the effectiveness of our proposed visual odometry technique on three datasets. These include two well-known and reliable sources: the KITTI visual odometry dataset and the Malaga Urban Dataset. We have also created our dataset, the custom Lunar Dataset. We only used sequences 11-21 from the KITTI dataset to train our depth estimation model, and we did not do any fine-tuning in the subsequent experimental sessions. In other words, we used Malaga Urban Dataset and custom Lunar Dataset just for validation and testing. If you want to stabilize it in indoor scenarios or other more complex environments, this will require you to retrain the depth network or fine-tune it under the type weights of our model. Furthermore, ablation experiments were conducted to assess the extent to which the proposed methodology affects the final result.

A. EXPERIMENTAL RESULTS AND ANALYSIS ON KITTI DATASET

KITTI Odometry Dataset [39], published in 2012 by the Karlsruhe Institute of Technology and Toyota Technological Institute in Chicago, is the definitive dataset for visual odometry research. The KITTI dataset encompasses 22 diverse real-world urban sequences. Of these, the initial 11 sequences are furnished with comprehensive data, including stereoscopic camera imagery, LiDAR point cloud scans, and ground truth, serving as a robust foundation for evaluation. In order to critically evaluate the performance of GD-MVO, we performed a comprehensive evaluation of the proposed method using 00-10 sequences and compared the results with the state-of-the-art methods in the field.

The methods compared include the geometry-based methods VISO2 and ORB-SLAM2, the end-to-end deep learning methods TSformer-VO, SC-SfM-Learner and Depth-VO-Feat, and the hybrid method DF-VO, RAUM-VO, Cho and Kim [21].

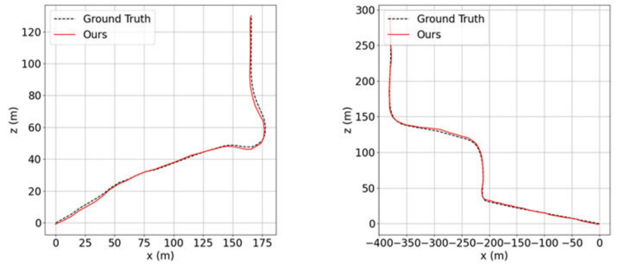


FIGURE 7. Malaga-urban-dataset-extract-02 (left) and Malaga-urban-dataset-extract-03 (right).

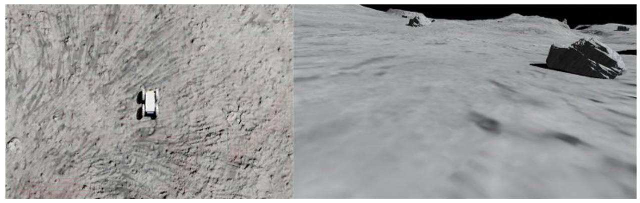


FIGURE 8. Overview of the scenarios for the lunar dataset.

In particular, since ORB-SLAM2 and SC-SfM-Learner suffer from scale ambiguity, we use the least squares method to adjust the scale factor appropriately to obtain the final camera trajectory, ensuring the fairness of the comparison. It is important to note that the literature [18] states that DF-VO is unable to yield accurate results for sequence 01. Consequently, they excluded this sequence from the calculation of the mean error. However, our experiments did not take this particular treatment.

A series of conventional metrics were employed to conduct a comprehensive assessment of the accuracy of the algorithm, as outlined below:

- Mean Translation Error(t_{err}) and Mean Rotation Error (r_{err}): These are the standardized metrics for the official KITTI odometer assessment.
- Absolute Translation Error(ATE): Estimate the root mean square of the Euclidean distance between each location point in the trajectory and the corresponding true trajectory location point.
- Relative Position Error(RPE): Measure the average relative position error between consecutive frames, including translation RPE(m) and rotation RPE(°).

The experimental results, encompassing both quantitative and qualitative analyses, are presented in Table 1 and Fig. 6. Fig. 6b demonstrates the unique ability of our method to operate reliably in highly dynamic environments, whereas competing methods such as DF-VO and ORB-SLAM2 have all failed under such challenging conditions. This is mainly because sequence 01 is a high-speed scene in which a large number of vehicles are traveling at high speeds. This makes it impossible for classical algorithms to track features along the frames, and end-to-end deep learning methods are unable to cope with this complex scenario at this stage.

TABLE 1. Quantitative results for the KITTI odometry dataset sequences 00-10. Optimal results are highlighted in red.

Method	Metric	00	01	02	03	04	05	06	07	08	09	10	Avg. Err.
ORB-SLAM2[6]	t_{err}	2.427	95.831	4.312	1.148	1.575	1.881	5.140	1.905	9.671	2.866	3.902	11.878
	r_{err}	0.350	0.446	0.311	0.187	0.268	0.198	0.230	0.281	0.296	0.249	0.305	**0.284**
	ATE	9.382	601.502	116.212	4.087	2.045	6.252	12.675	2.891	116.784	10.639	12.935	81.400
	RPE(m)	0.209	2.636	0.232	0.038	0.080	0.287	0.759	0.509	0.162	0.341	0.046	0.482
	RPE(°)	0.090	0.087	0.079	0.055	0.076	0.059	0.053	0.050	0.065	0.063	0.066	0.068
Depth-VO-Feat[12]	t_{err}	6.228	23.784	6.585	15.760	3.136	4.942	5.802	6.492	5.449	11.891	12.824	9.354
	r_{err}	2.441	1.753	2.264	10.621	2.018	2.343	2.062	3.558	2.388	3.604	3.411	3.315
	ATE	115.794	489.291	167.702	62.953	11.541	51.442	33.832	29.07	110.103	72.227	86.956	111.90
	RPE(m)	0.084	0.547	0.087	0.168	0.095	0.077	0.079	0.081	0.084	0.164	0.159	0.148
	RPE(°)	0.202	0.133	0.177	0.308	0.120	0.156	0.131	0.176	0.180	0.233	0.246	0.187
SC-SfM-Learner[11]	t_{err}	8.383	24.672	6.175	9.018	4.193	6.517	7.431	7.837	17.508	8.237	10.699	10.061
	r_{err}	3.394	1.314	1.960	4.929	2.012	2.379	1.646	4.531	2.611	2.193	4.577	2.868
	ATE	122.800	166.676	95.246	29.134	13.069	54.638	31.237	34.481	161.26	63.038	68.171	76.341
	RPE(m)	0.077	0.791	0.077	0.059	0.073	0.067	0.103	0.047	0.212	0.104	0.105	0.156
	RPE(°)	0.129	0.075	0.087	0.068	0.055	0.069	0.066	0.074	0.074	0.102	0.107	0.082
TSformer-VO-2[14]	t_{err}	5.152	33.400	4.870	14.444	6.854	10.735	17.703	23.205	4.146	8.091	13.713	12.938
	r_{err}	2.169	6.251	1.415	6.129	3.556	4.002	5.622	9.992	1.624	1.973	5.111	4.349
	ATE	50.066	209.038	62.188	14.737	4.244	54.688	50.519	36.059	28.534	41.126	21.131	52.03
	RPE(m)	0.028	0.751	0.053	0.128	0.083	0.144	0.209	0.143	0.032	0.092	0.145	0.164
	RPE(°)	0.364	0.751	0.053	0.128	0.083	0.144	0.209	0.143	0.032	0.092	0.145	0.195
DF-VO[18]	t_{err}	1.963	56.764	2.377	2.490	1.029	1.103	1.030	0.966	1.599	2.607	2.293	6.747
	r_{err}	0.598	13.930	0.548	0.391	0.251	0.303	0.304	0.269	0.318	0.288	0.369	1.597
	ATE	19.063	893.967	41.956	10.966	2.247	8.017	5.596	3.462	17.280	17.919	9.035	93.592
	RPE(m)	0.027	1.203	0.033	0.023	0.036	0.020	0.024	0.018	0.032	0.056	0.047	0.138
	RPE(°)	0.055	0.773	0.046	0.037	0.030	0.035	0.029	0.030	0.037	0.037	0.043	0.105
RAUM-VO [20]	t_{err}	2.548	8.354	2.578	3.217	2.860	3.045	3.033	2.390	3.636	2.927	5.843	3.676
	r_{err}	0.775	0.868	0.582	1.334	0.645	1.153	0.837	1.037	1.074	0.318	0.683	0.846
	ATE	16.272	23.748	16.139	2.602	2.283	17.470	9.234	2.164	16.303	8.664	12.297	**11.561**
	RPE(m)	0.040	0.257	0.050	0.030	0.052	0.038	0.046	0.028	0.053	0.068	0.078	0.067
	RPE(°)	0.059	0.062	0.048	0.048	0.035	0.044	0.042	0.058	0.045	0.042	0.051	0.049
Cho and Kim[21]	t_{err}	1.77	64.38	2.62	3.06	0.65	1.31	1.6	1.06	2.28	2.66	2.95	7.667
	r_{err}	0.79	16.87	0.74	0.89	0.55	0.74	0.56	0.67	0.76	0.53	0.95	2.186
	RPE(m)	0.03	1.71	0.04	0.02	0.04	0.02	0.03	0.02	0.03	0.06	0.05	0.186
	RPE(°)	0.06	0.67	0.06	0.04	0.04	0.05	0.05	0.04	0.05	0.05	0.06	0.106
Ours (ResNet 50)	t_{err}	1.581	2.517	1.121	2.185	3.513	1.944	0.995	1.530	1.401	1.394	1.380	**1.778**
	r_{err}	0.394	0.293	0.270	0.371	0.161	0.367	0.267	0.924	0.321	0.263	0.321	0.359
	ATE	21.919	34.203	14.949	8.526	9.560	8.228	5.713	5.193	21.995	11.901	9.068	13.750
	RPE(m)	0.038	0.128	0.031	0.038	0.077	0.036	0.027	0.029	0.033	0.028	0.026	**0.045**
	RPE(°)	0.063	0.045	0.051	0.049	0.037	0.044	0.037	0.047	0.045	0.046	0.052	**0.047**

TABLE 2. Absolute keyframe trajectory mean translation error on the KITTI dataset. All results except ours are taken from the original papers of the corresponding methods.

Sequence	TSSM-VO[13]	VIFT[15]	Ours
00	×	×	1.581
01	×	×	2.517
02	×	×	1.121
03	×	×	2.185
04	×	×	3.513
05	×	1.93	1.944
06	×	×	0.995
07	×	1.55	1.530
08	×	×	1.401
09	4.49	×	1.394
10	7.82	2.57	1.380

In addition, we have done a performance comparison for the latest Transform-based methods, as shown in Table 2 (including TSformer-VO in Table 1). It can be seen that the Transformer-based monocular visual odometry is far inferior to ours. Even facing the VIFT which incorporates IMU information, our method is still not inferior.

It is noteworthy that ORB-SLAM2 exhibits a considerable degree of drift in long-distance trajectory estimation in the absence of route loops (Fig. 6c and Fig. 6i). Although the KITTI dataset contains a variety of complex and challenging scenarios, our proposed method still exhibits excellent capabilities in trajectory estimation, with the best fit with ground truth trajectories.

B. EXPERIMENTAL RESULTS AND ANALYSIS ON MALAGA URBAN DATASET

The Malaga Urban Dataset [40], introduced by the University of Malaga's Robotics Laboratory in Spain in 2014, was designed to provide high-quality test data for SLAM, perception, and related technologies. This comprehensive urban dataset was collected using a sensor-equipped vehicle, featuring a Bumblebee2 stereo camera capturing high-resolution stereoscopic images at 20 frames per second (fps), alongside five LiDAR units. The data collection covered approximately 36 kilometers through Malaga's city center, encompassing intersections, pavements, and other typical urban road elements. Owing to its richness and specialized nature, this dataset has gained widespread adoption for evaluating visual odometry systems.

Given that our GD-MVO is a monocular visual odometry system, we opted to exclusively utilize the left-camera image sequence from the Malaga Urban Dataset for our experimental evaluations. As evident in Fig. 7, the method predicted trajectories closely align with the ground truth, underscoring both the robust generalization capability of the depth network and the broad applicability of the GD-MVO approach.

C. EXPERIMENTAL RESULTS AND ANALYSIS ON LUNAR DATASET

The motivation for creating the lunar dataset was to validate the adaptability and robustness of the proposed visual odometry approach in extreme environments with few textures and weak geometric features. This is of great significance for the future application of the algorithm in lunar rover and other space exploration missions.

In order to simulate a realistic lunar scene, we refer to images of the lunar surface taken by lunar probes and use webots software to build a high-fidelity 3D model of the moon. The model has an area of 50 m × 50 m and contains rich terrain details. Within this simulated lunar terrain, we deployed a four-wheel differential drive rover as our primary data collection platform. This rover, approximately 400 × 300 × 200 millimeters in size, is equipped with two monocular RGB cameras, as shown in Fig. 8. The simulation operates on a 32-millisecond time step, with RGB data captured at 960-millisecond intervals. Throughout its pseudo-randomized traversal of the simulated lunar terrain, the rover amassed a comprehensive dataset comprising approximately 2000 image frames, captured via its front-mounted RGB camera.

Despite the limited scope of this dataset, its low-texture scenarios closely mimic the visual complexities encountered in actual lunar exploration. Consequently, it provides an invaluable testbed for assessing the accuracy and robustness of visual odometry algorithms under conditions that push the boundaries of traditional computer vision techniques.

Fig. 9 showcases the trajectory prediction results of our proposed method on the lunar simulation dataset. It can be clearly seen that our method still has obvious shortcomings when dealing with similar low-textured scenes, as it is unable to accurately match features with repetitive structures in the scene. The trajectory in the figure appears to be rotated, which is due to the fact that the cart under the real path makes a purely rotational motion here, and our method is currently unable to handle similar degenerate motions.

Although we employ a realistic simulation, some disparities with the actual lunar surface remain. Moreover, the camera parameters in the simulation deviate from those encountered in real-world scenarios. Despite these various sources of potential error, our method exhibits considerable promise for robust visual odometry in lunar terrain.

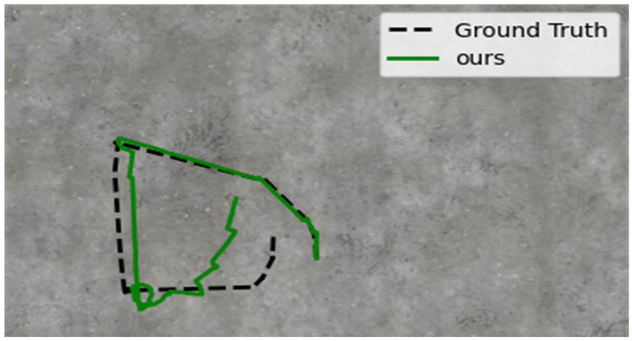


FIGURE 9. Trajectory in lunar dataset.

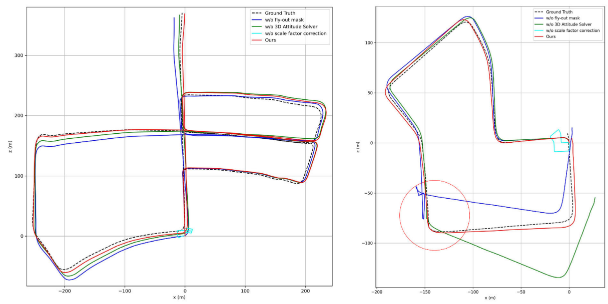


FIGURE 10. Trajectory of the ablation experiment in sequence 05 (left) and sequence 07(right).

TABLE 3. Ablation study.

	KITTI Seq.05		KITTI Seq.07	
	t_{err}	r_{err}	t_{err}	r_{err}
w/o fly-out mask	2.176	1.322	8.315	3.970
w/o 3D Attitude Solver	2.244	0.507	7.798	4.926
w/o scale factor correction	54.839	0.401	55.519	0.984
Ours	1.944	0.376	1.530	0.924

D. ABLATION EXPERIMENT

If any of the modules of the 3D attitude solver, flyout mask, and scale factor correction are removed from the algorithm, the performance of the GD-MVO is significantly degraded. This highlights the critical role these innovations play in maintaining robust visual odometry in complex and changing environments.

Table 3 presents the experimental outcomes, clearly demonstrating that the integration of both the fly-out mask and 3D Attitude Solver leads to substantial error reduction.

A particularly illustrative example is found in KITTI sequence 07, which features an image sequence rich in dynamic objects, including numerous pedestrians and vehicles in relative motion to the camera. Fig. 10 highlights this challenging area, enclosed within a red circle.

V. CONCLUSION

This paper proposes GD-MVO, a monocular visual odometry method that fuses the robustness of traditional geometric models with the adaptability of deep learning frameworks. The GD-MVO approach is rooted in well-established geometric theories and overcomes the limitations of purely data-driven techniques. Innovations include the introduction of fly-out masks to solve the fly-out padding problem, and the development of 2D and 3D attitude solvers to cope with dynamic scenes. As a result, this approach significantly enhances the accuracy and reliability of visual odometry.

While our experiments demonstrate the robustness and accuracy of GD-MVO under a variety of datasets, we recognize that its computational efficiency and real-time performance need to be further evaluated and optimized, as well as the method's performance under extreme low-light or high-reflectance conditions deserves further investigation. These conditions pose unique challenges to visual odometry algorithms due to the lack of reliable visual features and the possibility of sensor saturation or glare. In low-light environments, reduced signal-to-noise ratio and diminished contrast can hinder accurate feature detection and matching. Similarly, highly reflective surfaces may introduce specular highlights and distortions that can interfere with depth estimation and attitude recovery processes. Although our approach employs techniques such as flyout masks and dual solver systems to enhance robustness, the extent to which these innovations can mitigate the effects of extreme lighting conditions remains an open question.

To address this limitation and provide a more comprehensive assessment of GD-MVO's real-world applicability, we propose the following steps for future research: 1. Conduct targeted experiments in controlled low-light and highly reflective environments to quantify the method's performance degradation under these conditions. 2. Investigate the integration of complementary sensor modalities, such as thermal imaging or active illumination, to augment the visual input and improve robustness in challenging lighting scenarios. 3. Optimize feature selection in sparse data using the PSO [41] method to address the struggle of SIFT features in low-texture lunar environments. 4. Evaluate and optimize the real-time performance of GD-MVO and explore deployment and optimization on embedded platforms.

REFERENCES

[1] X. Yin, X. Wang, X. Du, and Q. Chen, "Scale recovery for monocular visual odometry using depth estimated with deep convolutional neural fields," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 5871–5879, doi: 10.1109/ICCV.2017.625.

[2] M. He, C. Zhu, Q. Huang, B. Ren, and J. Liu, "A review of monocular visual odometry," *Vis. Comput.*, vol. 36, no. 5, pp. 1053–1065, May 2020, doi: 10.1007/s00371-019-01714-6.

[3] R. I. Hartley, "In defense of the eight-point algorithm," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 19, no. 6, pp. 580–593, Jun. 1997, doi: 10.1109/34.601246.

[4] D. Nister, "An efficient solution to the five-point relative pose problem," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 26, no. 6, pp. 756–770, Jun. 2004, doi: 10.1109/TPAMI.2004.17.

[5] V. Lepetit, F. Moreno-Noguer, and P. Fua, "EPnP: An accurate O(n) solution to the PnP problem," *Int. J. Comput. Vis.*, vol. 81, no. 2, pp. 155–166, Feb. 2009, doi: 10.1007/s11263-008-0152-6.

[6] R. Mur-Artal and J. D. Tardós, "ORB-SLAM2: An open-source SLAM system for monocular, stereo, and RGB-D cameras," *IEEE Trans. Robot.*, vol. 33, no. 5, pp. 1255–1262, Oct. 2017, doi: 10.1109/TRO.2017.2705103.

[7] J. Engel, V. Koltun, and D. Cremers, "Direct sparse odometry," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 3, pp. 611–625, Mar. 2018, doi: 10.1109/TPAMI.2017.2658577.

[8] P. Agrawal, J. Carreira, and J. Malik, "Learning to see by moving," in *Proc. IEEE International Conference Computer Vision (ICCV)*, Dec. 2015, pp. 37–45, doi: 10.1109/ICCV.2015.13.

[9] S. Wang, R. Clark, H. Wen, and N. Trigoni, "DeepVO: Towards end-to-end visual odometry with deep recurrent convolutional neural networks," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2017, pp. 2043–2050, doi: 10.1109/ICRA.2017.7989236.

[10] B. Ummenhofer, H. Zhou, J. Uhrig, N. Mayer, E. Ilg, A. Dosovitskiy, and T. Brox, "DeMoN: Depth and motion network for learning monocular stereo," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5622–5631, doi: 10.1109/CVPR.2017.596.

[11] J.-W. Bian, Z. Li, N. Wang, H. Zhan, C. Shen, M. Cheng, and I. Reid, "Unsupervised scale-consistent depth and ego-motion learning from monocular video," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2019, pp. 1–11. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2019/hash/6364d3f0f495b6ab9dcf8d3b5c6e0b01-Abstract.html

[12] A. Ranjan, V. Jampani, L. Balles, K. Kim, D. Sun, J. Wulff, and M. J. Black, "Competitive collaboration: Joint unsupervised learning of depth, camera motion, optical flow and motion segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 12232–12241, doi: 10.1109/CVPR.2019.01252.

[13] H. Zhao, X. Qiao, Y. Ma, and R. Tafazolli, "Transformer-based self-supervised monocular depth and visual odometry," *IEEE Sensors J.*, vol. 23, no. 2, pp. 1436–1446, Jan. 2023, doi: 10.1109/JSEN.2022.3227017.

[14] A. O. Françani and M. R. O. A. Maximo, "Transformer-based model for monocular visual odometry: A video understanding approach," *IEEE Access*, vol. 13, pp. 13959–13971, 2025, doi: 10.1109/ACCESS.2025.3531667.

[15] Y. B. Kurt, A. Akman, and A. A. Alatan, "Causal transformer for fusion and pose estimation in deep visual inertial odometry," 2024, *arXiv:2409.08769*.

[16] K. Tateno, N. Navab, and F. Tombari, "Distortion-aware convolutional filters for dense prediction in panoramic images," in *Proc. ECCV*, in Lecture Notes in Computer Science, V. Ferrari, M. Hebert, C. Sminchisescu, and Y. Weiss, Eds., Jan. 2018, pp. 732–750, doi: 10.1007/978-3-030-01270-0_43.

[17] S. Yan Loo, A. Jahani Amiri, S. Mashohor, S. Hong Tang, and H. Zhang, "CNN-SVO: Improving the mapping in semi-direct visual odometry using single-image depth prediction," 2018, *arXiv:1810.01011*.

[18] H. Zhan, C. S. Weerasekera, J.-W. Bian, R. Garg, and I. Reid, "DF-VO: What should be learnt for visual odometry?" 2021, *arXiv:2103.00933*.

[19] N. Yang, R. Wang, J. Stückler, and D. Cremers, "Deep virtual stereo odometry: Leveraging deep depth prediction for monocular direct sparse odometry," in *Proc. ECCV*, in Lecture Notes in Computer Science, V. Ferrari, M. Hebert, C. Sminchisescu, and Y. Weiss, Eds., Jan. 2018, pp. 835–852, doi: 10.1007/978-3-030-01237-3_50.

[20] C. Cimarelli, H. Bavle, J. L. Sanchez-Lopez, and H. Voos, "RAUM-VO: Rotational adjusted unsupervised monocular visual odometry," *Sensors*, vol. 22, no. 7, p. 2651, Mar. 2022, doi: 10.3390/s22072651.

[21] H. M. Cho and E. Kim, "Dynamic object-aware visual odometry (VO) estimation based on optical flow matching," *IEEE Access*, vol. 11, pp. 11642–11651, 2023, doi: 10.1109/ACCESS.2023.3241961.

[22] D. G. Lowe, "Object recognition from local scale-invariant features," in *Proc. 7th IEEE Int. Conf. Comput. Vis.*, vol. 2, Oct. 1999, pp. 1150–1157, doi: 10.1109/ICCV.1999.790410.

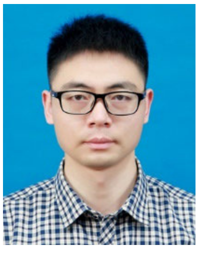
[23] G. L. Sicuranza and A. Carini, "A generalized FLANN filter for nonlinear active noise control," *IEEE Trans. Audio, Speech, Language Process.*, vol. 19, no. 8, pp. 2412–2417, Nov. 2011, doi: 10.1109/TASL.2011.2136336.

[24] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. MICCAI*, in Lecture Notes in Computer Science, N. Navab, J. Hornegger, W. M. Wells, and A. F. Frangi, Eds., Jan. 2015, pp. 234–241, doi: 10.1007/978-3-319-24574-4_28.
[25] C. Godard, O. M. Aodha, M. Firman, and G. Brostow, "Digging into self-supervised monocular depth estimation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3827–3837, doi: 10.1109/ICCV.2019.00393.
[26] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004, doi: 10.1109/TIP.2003.819861.
[27] M. Jaderberg, K. Simonyan, A. Zisserman, and k. Kavukcuoglu, "Spatial transformer networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 1–15. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2015/hash/33ceb07bf4eeb3da587e268d663aba1a-Abstract.html
[28] C. Godard, O. M. Aodha, and G. J. Brostow, "Unsupervised monocular depth estimation with left-right consistency," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6602–6611, doi: 10.1109/CVPR.2017.699.
[29] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," 2019, *arXiv:1912.01703*.
[30] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[31] R. Garg, V. K. Bg, G. Carneiro, and I. Reid, "Unsupervised CNN for single view depth estimation: Geometry to the rescue," 2016, *arXiv:1603.04992*.
[32] T. Zhou, M. Brown, N. Snavely, and D. G. Lowe, "Unsupervised learning of depth and ego-motion from video," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6612–6619, doi: 10.1109/CVPR.2017.700.
[33] H. Jiang, L. Ding, Z. Sun, and R. Huang, "DiPE: Deeper into photometric errors for unsupervised learning of depth and ego-motion from monocular videos," 2020, *arXiv:2003.01360*.
[34] R. Mahjourian, M. Wicke, and A. Angelova, "Unsupervised learning of depth and ego-motion from monocular video using 3D geometric constraints," 2018, *arXiv:1802.05522*.
[35] C. Kerl, J. Sturm, and D. Cremers, "Dense visual SLAM for RGB-D cameras," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Nov. 2013, pp. 2100–2106, doi: 10.1109/IROS.2013.6696650.
[36] D. P. Frost, O. Kähler, and D. W. Murray, "Object-aware bundle adjustment for correcting monocular scale drift," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2016, pp. 4770–4776, doi: 10.1109/ICRA.2016.7487680.
[37] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," *Commun. ACM*, vol. 24, no. 6, pp. 381–395, Jun. 1981, doi: 10.1145/358669.358692.
[38] G. Terzakis and M. Lourakis, "A consistently fast and globally optimal solution to the perspective-n-point problem," in *Proc. ECCV*, in Lecture Notes in Computer Science, A. Vedaldi, H. Bischof, T. Brox, and J.-M. Frahm, Eds., Jan. 2020, pp. 478–494, doi: 10.1007/978-3-030-58452-8_28.
[39] A. Geiger, P. Lenz, C. Stiller, and R. Urtasun, "Vision meets robotics: The KITTI dataset," *Int. J. Robot. Res.*, vol. 32, no. 11, pp. 1231–1237, Sep. 2013, doi: 10.1177/0278364913491297.
[40] J.-L. Blanco-Claraco, F.-Á. Moreno-Dueñas, and J. González-Jiménez, "The Málaga urban dataset: High-rate stereo and LiDAR in a realistic urban scenario," *Int. J. Robot. Res.*, vol. 33, no. 2, pp. 207–214, Feb. 2014, doi: 10.1177/0278364913507326.
[41] A. Amjad, L.-C. Tai, and H.-T. Chang, "Utilizing enhanced particle swarm optimization for feature selection in gender-emotion detection from english speech signals," *IEEE Access*, vol. 12, pp. 189564–189573, 2024, doi: 10.1109/ACCESS.2024.3516790.

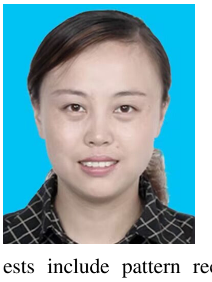
ZHUANG LI received the B.S. degree in information and computing science from Henan Polytechnic University, Jiaozuo, China, in 2021. He is currently pursuing the M.E. degree with the School of Computer Science, Jiangsu University of Science and Technology, Zhenjiang, China. He is engaged in research on computer vision and its application.

JIANJUN CHEN received the B.S. degree in applied mathematics from Shanxi Datong University, Datong, China, in 2007, the M.S. degree in applied mathematics from Dalian Jiaotong University, Dalian, China, in 2010, and the Ph.D. degree in intelligent systems design engineering from Toyama Prefectural University, Toyama, Japan, in 2015. He is currently engaged in research on soft computing and its application to assistive systems for people with disabilities.

ABDUL WAHEED received the B.Sc. degree in double computer science from The Islamia University of Bahawalpur, Pakistan, in 2019, and the master's degree in computer science from Lahore Garrison University, Lahore, Pakistan, in 2021. He is currently engaged in a research project on the application of artificial intelligence with Dr. Prof. Jianjun Chen at Jiangsu University of Science and Technology, Zhenjiang, China. During the master's degree, he received the Gold Medal.



FEI WANG received the Ph.D. degree in computer software and theory from Wuhan University, China, in 2020. He has published multiple academic papers in international conferences and leading journals, including WAIM, *Frontiers of Computer Science*, *Journal of Software*, and *Journal of Computer Research and Development*. His current research interests include patent mining, information retrieval, knowledge graph, and natural language processing.



YAN FAN received the B.S. degree in computer science from East China Shipbuilding Institute, Zhenjiang, China, in 2001, and the M.S. degree in computer science from Jiangsu University of Science and Technology, Zhenjiang, in 2004. She accepted continuing education in SUPINFO, Paris, France, in August 2005. She is currently an Associate Professor with the School of Computer Science, Jiangsu University of Science and Technology, Zhenjiang. Her current research interests include pattern recognition, feature extraction, and deep learning applications.